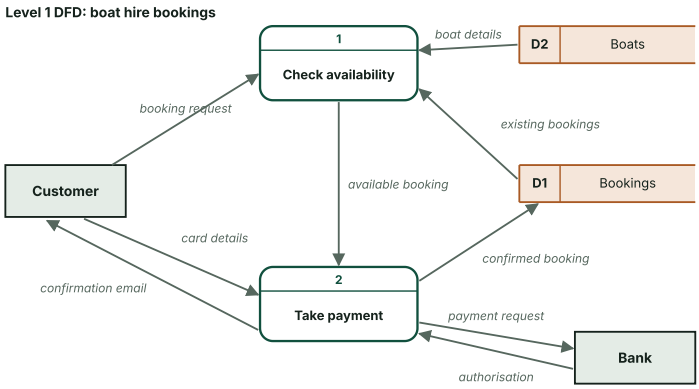
Data flow diagrams

AP05
FACT FILE

Specification: students should be able to

- Explain the purpose of a data flow diagram (DFD) and its advantages
- Identify and use the standard DFD elements: external entities (sources and sinks), processes, data stores and data flows
- Produce a context diagram (Level 0 DFD) for a simple scenario
- Produce a Level 1 DFD by exploding a context diagram, maintaining consistency between levels
- Interpret DFDs and identify and correct errors using the rules for valid DFDs

Key words: data flow diagram (DFD), context diagram, Level 0, Level 1, external entity, source, sink, process, data store, data flow, system boundary, granularity, explode, decomposition, balancing, duplicated entity



A level 1 data flow diagram for a boat hire booking system with external entities, numbered processes, data stores and labelled data flows

Modelling a system with data flows

A data flow diagram, or DFD, is a graphical model showing how data moves through a system. The idea is simple: data arrives from outside the system, processes act on it, some of it is stored for later, and new data flows back out to people or organisations. DFDs are widely used during analysis because they capture the analyst's understanding of the existing system, and of the proposed one, in a form that is easy to check. They rely on only a handful of symbols, so a client or end user with little technical knowledge can follow the diagram and point out anything that is missing or wrong. That builds an agreed, common understanding between the analyst, the client and the users before design begins. DFDs can be drawn at different levels of detail, called granularity: a coarse, high-level overview called a context diagram or Level 0 DFD, and more detailed Level 1 diagrams that show the main sub-processes and data stores. Carefully drawn DFDs can also reveal problems in the current system, such as bottlenecks where too much data passes through one process, unnecessary duplication, or data that is stored but never used. A DFD shows what happens to data, not the order of steps or the decisions made.

The elements of a DFD

A DFD is built from a small set of elements. An external entity is a person, organisation or other system outside the system boundary that supplies data to the system, when it is called a source, or receives data from it, when it is called a sink. Many entities, such as a customer who sends an order and receives an invoice, are both. In the notation used in the CCEA textbook, external entities are drawn as ellipses; if the same entity has to appear twice to avoid crossing lines, it is drawn again with a short diagonal line to show that it is a duplicate. A process transforms incoming data into outgoing data and is drawn as a rectangle with a strip at the top holding its number, such as 1.0 or 2.0, and is named with a verb phrase such as 'Calculate invoice'. A data store, drawn as an open-ended rectangle with an identifier such as D1, holds data for later use, for example a stock file or a customer database table. A data flow is an arrow, labelled with the data it carries, showing the direction of movement. Other textbooks use squares for entities and rounded rectangles for processes; either notation is acceptable if it is applied consistently.

Context diagrams

The context diagram, also known as a Level 0 DFD, is the highest-level and least detailed view of a system. Its main job is to show the boundary of the system: what lies inside it and what lies outside. The entire system is represented as one single process, often numbered 0, in the centre of the diagram. Around it are all the external entities with which the system exchanges data, connected by labelled data flows that show what each entity sends to the system and receives from it. Detailed internal matters, such as checking a customer's credit rating or updating individual files, are deliberately left out at this level. Data stores are normally omitted, or used very sparingly, because the purpose is to define scope rather than to show how the system works internally. Because it gives such a clear overview, a context diagram is particularly useful for managers who need to see what the system will cover and which parts of the organisation it will interact with. Drawing it also forces the analyst and client to agree the scope: for example, whether purchasing goods from suppliers or carrying out marketing is part of a sales system, or outside it and therefore an external entity or ignored entirely.

Level 1 DFDs and balancing

A Level 1 DFD is produced by exploding, or decomposing, the single process in the context diagram into the main sub-processes that make up the system. Each sub-process is numbered, typically 1.0, 2.0, 3.0, and given a verb-phrase name. The data stores that these processes use are added, along with the internal data flows between processes and between processes and stores. The external entities do not change, because they are outside the system and are not being explored in more detail. The crucial rule linking the two levels is consistency, often called balancing. Every data flow that enters or leaves the system on the context diagram must also appear on the Level 1 diagram, connected to whichever sub-process now handles it, and with the same label. Nothing should cross the system boundary at Level 1 that was not shown at Level 0. If a context diagram shows a two-way exchange between a customer and the system, the Level 1 diagram must still show both flows. The same idea continues downwards: a Level 1 process can be exploded into a Level 2 diagram with processes numbered 2.1, 2.2 and so on, although CCEA AS candidates only need to produce context and Level 1 diagrams.

Rules for valid DFDs

Several rules make a DFD valid, and many exam questions ask you to apply them. First, data must always pass through a process: a flow cannot go directly from one external entity to another, from one data store to another, or between an external entity and a data store. If a customer's order needs to be stored, a process such as 'Record order' must sit between the customer and the orders file. Second, every process must have at least one input and at least one output, because its job is to transform data; a process with only inputs is sometimes called a black hole, and one with only outputs is a miracle. Third, every data store should be written to by at least one process and read by at least one process somewhere in the system, otherwise it serves no purpose. Fourth, every data flow must be labelled with a noun phrase that names the data it carries, such as 'Payment details', rather than an action. Processes are named with verb phrases, and both processes and data stores are given unique identifiers. Finally, levels must balance. Checking a finished diagram against each of these rules in turn is a reliable way to find mistakes before submitting it.

Drawing a DFD from a scenario

Turning a written description into a DFD is a skill that improves with a methodical approach. Begin by reading the scenario twice. Underline the people, departments and organisations that send data into the system or receive data from it; these are your candidate external entities. Then look for verbs describing actions on data, such as receives, checks, calculates, records or sends, which suggest processes, and for nouns describing records that are kept, such as a stock file or member list, which suggest data stores. Draw the context diagram first, showing one process for the whole system, all the entities and every labelled flow between them. Next, explode that process into two to four main sub-processes, decide which stores each one reads or writes, and add the internal flows. Finally, check the Level 1 diagram against the context diagram for balancing and against each validity rule. Remember that departments inside the organisation being modelled may be processes rather than entities, depending on where the boundary is drawn, so read carefully what the question says the system covers. Using the scenario's own names for entities and data makes the diagram easier to mark and avoids ambiguity.

T Key terms from the mark schemes

Data flow diagram (DFD)
A diagram that models how data moves through a system using external entities, processes, data stores and labelled data flows.

Context diagram
A top-level (Level 0) DFD showing the whole system as a single process with its external entities and the data flows between them, defining the system boundary.
Credited: Level 0 DFD

Level 1 DFD
A DFD in which the single process of the context diagram is exploded into its main numbered sub-processes, with data stores and internal flows added.

External entity
A person, organisation or system outside the system boundary that is a source or destination of data.
Credited: Source/sink; terminator

Source
An external entity from which data enters the system.

Sink
An external entity to which the system sends data.
Credited: Destination

Process
An activity in a DFD that transforms input data into output data; named with a verb phrase and numbered.

Data store
A place in a DFD where data is held for later use, such as a file or database table.
Credited: File

Data flow
A labelled arrow in a DFD showing data moving between entities, processes and data stores.

Balancing
The rule that the data flows into and out of a process (or the whole system) must be the same on its exploded lower-level DFD.
Credited: Consistency between levels

Explode
To break a DFD process down into more detailed sub-processes on a lower-level diagram.
Credited: Decompose

Exam focus

DFD questions range from one-mark identification to drawing a full diagram. When asked for the purpose of a DFD, mark schemes credit points such as providing a visual representation of the system, showing it at different levels of detail, showing how the system interacts with external entities and identifying the processes and data stores. You may be shown a diagram with parts labelled A to D and asked to identify the external entity, data flow, process and data store, so learn the shapes. When drawing a context diagram, marks are typically awarded for a single correctly named process, each correct external entity and each correctly labelled flow in the right direction; unlabelled arrows usually lose marks. For a Level 1 diagram, show numbered processes, the data stores and flows, and make sure it balances with the context diagram. If asked about the relationship between the two levels, explain that Level 1 explodes the single process into sub-processes, adds data stores and keeps the same external flows. When identifying an error, name the rule broken and give a specific correction. Common mistakes are drawing a flowchart with decisions, connecting entities directly to stores, and omitting flow labels.

? Check yourself

1. An analyst plans to show a Newtownabbey car dealership's managers a set of data flow diagrams of their proposed servicing system. Describe two ways the diagrams will be useful. [2]
2. In a DFD for a school library's loans system, explain how a source differs from a sink, giving an example of each from the library. [4]
3. A trainee analyst at a Lisburn software firm has drawn a context diagram for a car-wash booking app and is about to draw the Level 1 DFD. State two ways in which the Level 1 diagram will differ from the context diagram. [2]

Answers: 1. Any two from: provides a visual/graphical representation of the system (1) / shows the system at different levels of detail (1) / shows how the system interacts with external entities / defines the boundary (1) / identifies the processes and data stores (1) / gives a shared understanding that users can check (1) / helps identify bottlenecks or problems in the current system (1) 2. A source lies outside the system and sends data into it (1), e.g. a pupil/borrower sending a loan request or book return (1) / a sink lies outside the system and is sent data by it (1), e.g. a pupil receiving an overdue reminder, or a librarian/teacher receiving a report (1) 3. Any two from: context diagram has a single process; Level 1 has several numbered sub-processes (1) / context diagram normally shows no data stores; Level 1 shows data stores (1) / Level 1 shows internal data flows between processes and stores; context shows only flows crossing the boundary (1) / context gives an overview for management; Level 1 gives more detail (1)